

Problem. Download the following dataset: [Link to the dataset](#).

Dataset context: Car drivers like to adjust the seat position for their own comfort. Car designers would find it helpful to know where different drivers will position the seat depending on their size and age. Researchers at the HuMoSim laboratory at the University of Michigan collected data on 38 drivers. The dataset contains the following variables:

- **Age:** Age in years.
- **Weight:** Weight in lbs.
- **HtShoes:** Height in shoes in cm.
- **Ht:** Height bare foot in cm.
- **Seated:** Seated height in cm.
- **Arm:** Lower arm length in cm.
- **Thigh:** Thigh length in cm.
- **Leg:** Lower leg length in cm.
- **hipcenter:** Horizontal distance of the midpoint of the hips from a fixed location in the car in mm.

- (a) (*2pts*) Read the dataset, and then perform a full regression model:

$$\text{hipcenter} = \beta_0 + \beta_1 \text{Age} + \beta_2 \text{Weight} + \dots + \beta_8 \text{Leg} + \epsilon, \epsilon \sim \mathcal{N}(0, \sigma^2)$$

Interpret the estimated coefficients, and perform residual analysis.

- (b) (*1pt*) Compare this model to the noise-only model:

$$\text{hipcenter} = \beta_0 + \epsilon, \epsilon \sim \mathcal{N}(0, \sigma^2)$$

Use an appropriate statistical test with significant level $\alpha = 0.05$. Comment on the result.

- (c) (*2pts*) Use the `glmnet` library to set up a full ridge regression model. Perform cross validation to choose the best λ , you may use the search space $\lambda \in \text{seq}(1\text{e-}3, 50, 1\text{e-}3)$. Comment on the result.
- (d) (*1pt*) With the optimal ridge model, predict the `hipcenter` values using the data points from the dataset `newdata.csv`: [Link to this dataset](#).
- (e) (*3pts*) Use the `glmnet` library to set up a full LASSO regression model. Perform cross validation to choose the best λ , you may use the search space $\lambda \in \text{seq}(1\text{e-}3, 50, 1\text{e-}3)$.
With the optimal λ , comment on the estimated coefficients. Which variables remains non-zero ?
Compare the estimated coefficients and its corresponding OLS model.
- (f) (*1pt*) Using the dataset `newdata.csv`, compare the predicted values between the ridge model and the LASSO model.

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